

01286326 THEORY OF COMPUTATION
ASSIGNMENT 2

PROBLEM 1

Suppose M is a finite automaton $(Q, \Sigma, \delta, q_0, F)$ where

- $Q = \{q_0, q_1, q_2, q_3, q_4\}$
- $\Sigma = \{0,1\}$
- δ is described as

	0	1
q_0	q_1	q_3
q_1	q_2	q_1
q_2	q_4	q_1
q_3	q_3	q_4
q_4	q_0	q_2

- q_0 is the initial state,
- $F = \{q_1, q_4\}$

Show how to construct a regular expression α equivalent to this automaton, i.e. $L(\alpha) = L(M)$.

PROBLEM 2

Let β be the following regular expression over the alphabet $\Sigma = \{0,1\}$:

$$\beta = (01 \cup 011)^+(10 \cup 110)^*$$

Show how to construct a nondeterministic finite automaton N equivalent to this regular expression, i.e. $L(N) = L(\beta)$.

PROBLEM 3

Let R be the following language over the alphabet $\Sigma = \{0,1\}$:

$$R = L((0 \cup 10)^*) - L((\Sigma\Sigma)^*)$$

Find a regular expression φ such that $L(\varphi) = R$.

PROBLEM 4

For each language below, prove or disprove whether it is regular. Here, $\Sigma = \{0,1\}$.

- (1) $L_1 = \{w \in \Sigma^* \mid w \text{ is a palindrome}\}$. Note that a *palindrome* is a string that reads the same forward and backward, such as 10101 and 0110.
- (2) $L_2 = \{0^m 1^n \mid m \geq n \geq 1 \text{ and } m \text{ is divisible by } n\}$.
- (3) $L_3 = \{ww \mid w \in \Sigma^*\} \cap \{w \in \Sigma^* \mid w \text{ contains no consecutive 0s and no consecutive 1s}\}$.

PROBLEM 1

Suppose M is a finite automaton $(Q, \Sigma, \delta, q_0, F)$ where

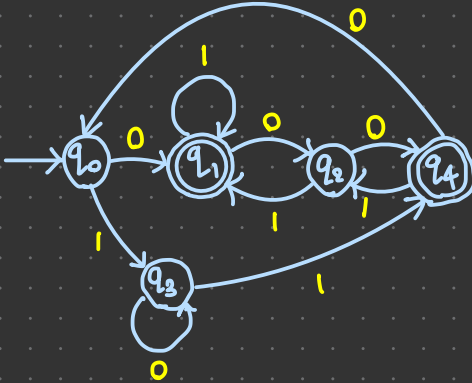
- $Q = \{q_0, q_1, q_2, q_3, q_4\}$
- $\Sigma = \{0, 1\}$
- δ is described as

	0	1
q_0	q_1	q_3
q_1	q_2	q_1
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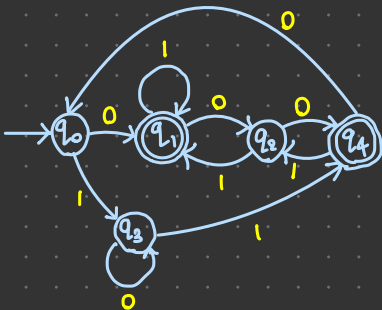
- q_0 is the initial state,
- $F = \{q_1, q_4\}$

Show how to construct a regular expression α equivalent to this automaton, i.e. $L(\alpha) = L(M)$.

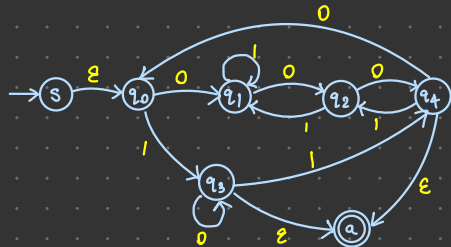
DFA



Since every DFA is an NFA, we already got NFA and will convert it into GNFA first.

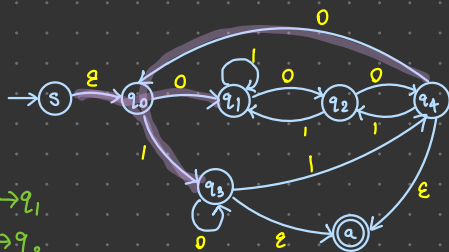


NFA



GNFA

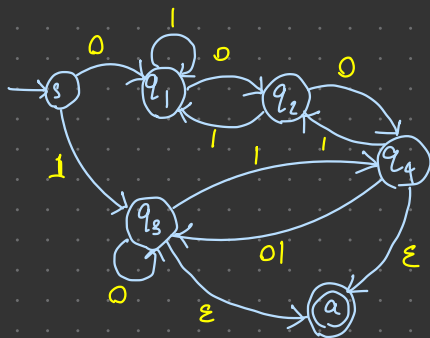
Removing q_0



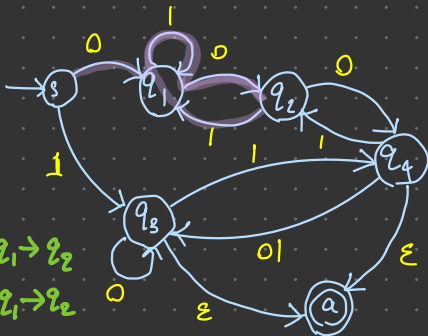
$s \rightarrow q_0 \rightarrow q_1$

$s \rightarrow q_0 \rightarrow q_3$

$q_4 \rightarrow q_0 \rightarrow q_3$

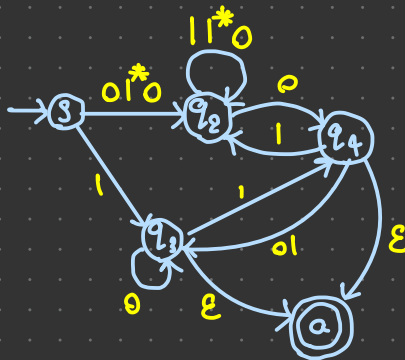


Removing q_1

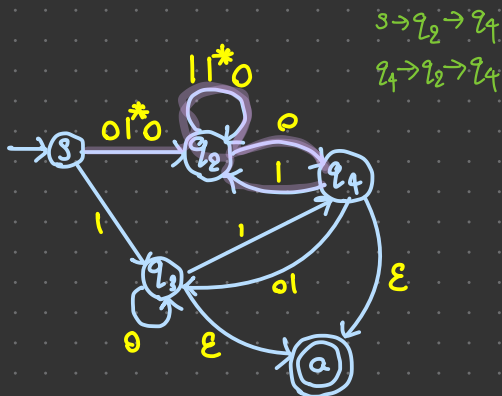


$s \rightarrow q_1 \rightarrow q_2$

$q_2 \rightarrow q_1 \rightarrow q_2$

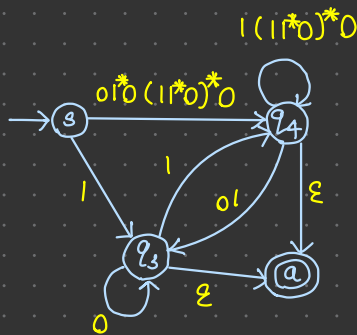


Removing q_2

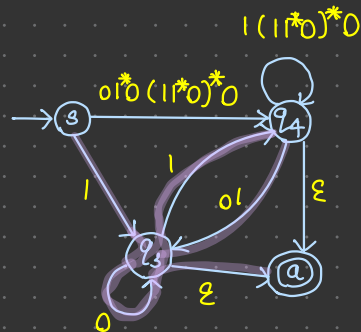


$s \rightarrow q_2 \rightarrow q_4$

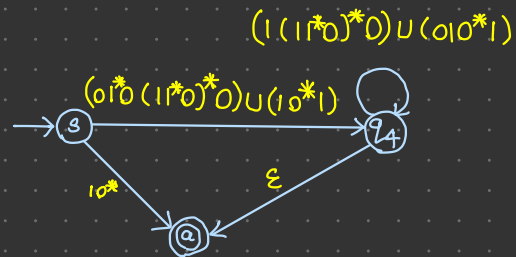
$q_4 \rightarrow q_2 \rightarrow q_4$



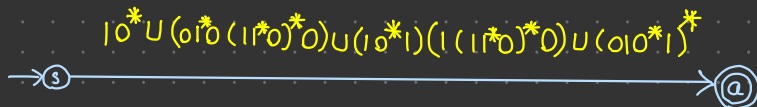
Removing q_3



$s \rightarrow q_3 \rightarrow q_4$
 $s \rightarrow q_3 \rightarrow a$
 $q_4 \rightarrow q_3 \rightarrow q_4$



Removing q_4



PROBLEM 2

Let β be the following regular expression over the alphabet $\Sigma = \{0,1\}$:

$$\beta = (01 \cup 011)^+(10 \cup 110)^*$$

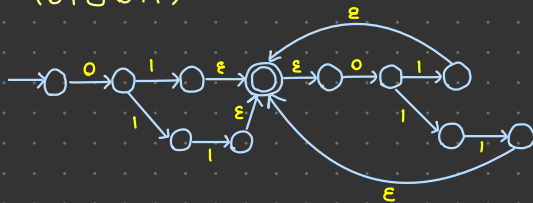
Show how to construct a nondeterministic finite automaton N equivalent to this regular expression, i.e. $L(N) = L(\beta)$.

*	Matches a sequence of zero or more matches of the preceding atom
+	Matches a sequence of one or more matches of the preceding atom

$a^* = \{\epsilon, a, aa, aaa, \dots\}$

$a^+ = \{a, aa, aaa, \dots\}$

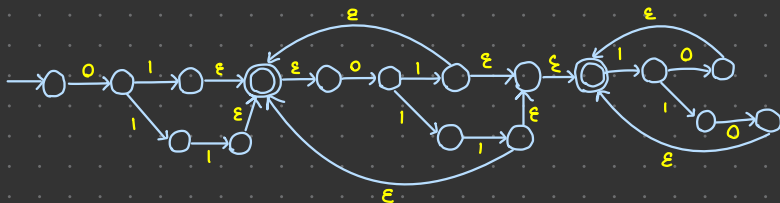
$(01 \cup 011)^+$



$(10 \cup 110)^*$



N



PROBLEM 3

Let R be the following language over the alphabet $\Sigma = \{0,1\}$:

$$R = L((0 \cup 10)^*) - L((\Sigma\Sigma)^*)$$

Find a regular expression φ such that $L(\varphi) = R$.

$$L((0 \cup 10)^*) = \{\epsilon, 0, 10, 00, 010, 100, 1010, \dots\}$$

$$L((\Sigma\Sigma)^*) = \{\epsilon, 00, 01, 10, 11, 0000, 0101, 1010, 0110, 1001, \dots\}$$

↑

strings of even length where every pair of consecutive bits can be anything from $\Sigma = \{0,1\}$

$$R = L((0 \cup 10)^*) - L((\Sigma\Sigma)^*)$$

↳ all strings from $L((0 \cup 10)^*)$ except those are even-length strings (odd lengths)

$$\{\cancel{\epsilon}, \cancel{0}, \cancel{10}, \cancel{00}, \cancel{010}, \cancel{100}, \cancel{1010}, \dots\}$$

$$\{\cancel{\epsilon}, \cancel{00}, \cancel{01}, \cancel{10}, \cancel{11}, \cancel{0000}, \cancel{0101}, \cancel{1010}, \cancel{0110}, \cancel{1001}, \dots\}$$

$$L(\emptyset) = (0 \cup 10)((0 \cup 10)(0 \cup 10))^+ \quad \text{not sure!}$$

PROBLEM 4

For each language below prove or disprove whether it is regular. Here, $\Sigma = \{0,1\}$.

- (1) $L_1 = \{w \in \Sigma^* \mid w \text{ is a palindrome}\}$. Note that a *palindrome* is a string that reads the same forward and backward, such as 10101 and 0110.
- (2) $L_2 = \{0^m 1^n \mid m \geq n \geq 1 \text{ and } m \text{ is divisible by } n\}$.
- (3) $L_3 = \{ww \mid w \in \Sigma^*\} \cap \{w \in \Sigma^* \mid w \text{ contains no consecutive 0s and no consecutive 1s}\}$.

$$1) L_1 = \{w \in \Sigma^* \mid w \text{ is a palindrome}\}$$

Proof:

Assume L_1 is a regular language.

By Pumping Lemma, there exists a pumping length p .

Assume a string from L_1 , $s = 0^p 11 0^p$

By decomposing string s into x, y and z ,

$$\left. \begin{array}{l} x = 0^a, a \geq 0 \text{ since } |x| \text{ can be empty} \\ y = 0^b, b \geq 1 \text{ since } |y| > 0 \end{array} \right\} |xy| \leq p$$

$$z = 0^{p-d-R} 11 0^p$$

According to pumping lemma, $xy^iz \in L_1$ for each $i \geq 0$

$$\text{for } i, y^i = y^i R$$

$$xy^iz \Rightarrow 0^{iR} 0^{p-d-R} 11 0^p$$

$$\Rightarrow 0^{p+iR-R} 11 0^p$$

should be equal if it is palindrome.

According to language L_1 which are all strings of palindrome,

$$p+iR-R = p$$

$i=1$, which means only when $i=1$, it will make our string a palindrome.

Suppose, we choose $i=2$,

$$xy^iz \Rightarrow 0^{p+R} 11 0^p$$

There are more 0s in first p , so it contradicts!

Therefore L_1 is not a language.

2) $L_2 = \{0^m 1^n \mid m \geq n \geq 1 \text{ and } m \text{ is divisible by } n\}$

Proof:

Assume L_2 is a regular language.

By Pumping Lemma, there exists a pumping length p .

Assume a string from L_2 , $s = 0^{2p} 0^p$

By decomposing string s into x, y and z ,

$$\begin{aligned} x &= 0^d, d \geq 0 \text{ since } x \text{ can be empty} \\ y &= 0^R, R \geq 1 \text{ since } |y| > 0 \end{aligned} \quad \left. \vphantom{\begin{aligned} x &= 0^d, d \geq 0 \text{ since } x \text{ can be empty} \\ y &= 0^R, R \geq 1 \text{ since } |y| > 0 \end{aligned}} \right\} |xy| \leq p$$

$$z = 0^{2p-d-R} 1^p$$

According to pumping lemma, $xy^iz \in L_2$ for each $i \geq 0$

$$\text{for } i, y^i = y^i R$$

$$xy^iz \Rightarrow 0^{iR} 0^{2p-d-R} 1^p$$

$$\Rightarrow 0^{2p+iR-R} 1^p$$

Suppose: $i=2$,

$$xy^iz \Rightarrow 0^{2p+R} 1^p$$

According to language L_2 which are all strings having * of 0s is divisible by * of 1s,

$2p+R$ should be divisible by p and $1 \leq R \leq p$,

which is not true! $2p+R$ cannot be divisible by p . A contradiction!

Therefore, L_2 is not regular.

(3) $L_3 = \{ww \mid w \in \Sigma^*\} \cap \{w \in \Sigma^* \mid w \text{ contains no consecutive 0s and no consecutive 1s}\}$.

$\{00, 11, 0101, 001001, \dots\} \cap \{0, 1, 01, 10, 010, 101, 0101, 1010, \dots\}$
no consecutive 0s and 1s with the same number of 0s and 1s.

Proof:

Assume L_3 is a regular language.

By Pumping Lemma, there exists a pumping length p .

Assume a string from L_3 , $s = (01)^p (01)^p$

By decomposing string s into x, y and z ,

$$\left. \begin{aligned} x &= (01)^\alpha, \alpha \geq 0 \text{ since 'x' can be empty} \\ y &= (01)^\beta, \beta \geq 1 \text{ since } |y| > 0 \\ z &= (01)^{p-\alpha-\beta} (01)^p \end{aligned} \right\} |xyz| \leq p$$

According to pumping lemma, $xy^i z \in L_3$ for each $i \geq 0$

for i , $y^i = y^i \beta$

$$\begin{aligned} xy^i z &\Rightarrow (01)^\alpha (01)^{i\beta} (01)^{p-\alpha-\beta} (01)^p \\ &\Rightarrow (01)^{p+i\beta-\beta} (01)^p \end{aligned}$$

According to language L_3 which contains strings that can be written as ww , where w does not contain consecutive 0s and consecutive 1s.

$p+i\beta-\beta$ should be equal to p

$$\text{but } p+i\beta-\beta \neq p \\ i \neq 1$$

Suppose we choose $i=2$,

$$xy^i z \Rightarrow (01)^{p+\beta} (01)^p \text{ which is out of our pattern: } (01)^p (01)^p$$

$1 \leq \beta \leq p$

A contradiction!

Therefore, L_3 is not regular.